Problem-Solution Fit

Date	12 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviours, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • **Students** • **Teachers** • **Educational Institutions** • **Online Learning Platforms** • **Training Centers**	**5. AVAILABLE SOLUTIONS (PROS & CONS) [AS]** **Existing Solutions:** • **Manual evaluation by teachers** • **Online quizzes** • **Keyword-based assessment systems** **Pros:** • **Easy to conduct** • **Widely used** **Cons:** • **Time-consuming** • **Subjective evaluation** • **Cannot understand semantic meaning**	**8. CHANNELS OF BEHAVIOR [CH]** **Online** • **Streamlit Web Application** • **GitHub Repository** • **Educational Platforms** • **Learning Management Systems** Offline • Classroom Demonstrations • Computer Laboratories • Workshops • Academic Presentations

	• **Limited personalized feedback**	
2. PROBLEMS / PAINS • Difficult to evaluate conceptual understanding through spoken explanations. • Manual assessment consumes significant time. • Traditional keyword-based evaluation lacks semantic understanding. • Students receive delayed and limited feedback. Frequency: High – Occurs during assignments, viva examinations, and online assessments.	**6. CUSTOMER LIMITATIONS (Budget, Devices) [CL]** • **Limited computing resources.** • **Internet availability may vary.** • **Users may have low-end laptops.** • **Different audio recording quality.**	**9. PROBLEM ROOT / CAUSE [RC]** • Lack of automated concept evaluation tools. • Traditional systems rely on keyword matching. • Manual assessment is slow and inconsistent. • Limited personalized feedback mechanisms.
3. TRIGGERS TO ACT [TR] • Need for faster and automated assessment. • Increasing adoption of AI in education. • Demand for objective concept evaluation. • Requirement for instant feedback and reports.	**7. BEHAVIOR + ITS INTENSITY [BE]** • Frequently uploads voice explanations. • Expects instant analysis and feedback. • Downloads reports for review. Intensity:	**10. YOUR SOLUTION [SL]** Voice-Based Concept Understanding Analyser (VBCUA) • Upload spoken explanation. • Convert speech to text using Faster Whisper. • Compare with reference concepts using Sentence-BERT.

	High during assignments, projects, and examinations.	• Perform keyword matching. • Generate AI-based scores and feedback. • Store results in SQLite database. • Generate downloadable PDF reports. • Display analytics through an interactive dashboard.
4. EMOTIONS Before: • Confused • Nervous • Uncertain about understanding After: • Confident • Motivated • Better understanding through AI feedback		